Final cleaning report

Team Name: SLU 0501 DVT Team 22

Team Members:
Abu Yussif (abuyussif125@gmail.com)
Adeniyi Aishat (aishatadeniyi418@gmail.com)
Ayesha Ashraf (ayeshaashraf015@gmail.com)
Mpfariseni Ephraim (ephraimmpfariseni@gmail.com)
Madhubala Khatri (madhubalakhatri53@gmail.com)

Introduction
This report outlines the final data cleaning process carried out on the dataset to prepare it for analysis.

Steps Taken
1.Deleted several columns and renamed some columns for better identification, analysis, and dashboard presentation.
2.The initial dataset contained 40 columns; after final cleaning, it now contains 27 columns.
3.A total of 13 columns were deleted.

Deleted Columns and Reasons

Column Name	Reason for Deletion
Received_at, Created_At, Modified_At, Created_At_2, Modified_At_2, Created_At_3, Modified_At_3	These are system-generated timestamps consisting of unique numbers with no analytical value.
University_1, University_2	Duplicates of the main University column and have no impact on analysis.
Reference_ID_2	Supposed to contain unique values but had missing entries and duplicates, which could interfere with the original Reference_ID.
ID	Contained no unique identifiers; therefore, not useful for analysis.
Degree_Type	Contained only one value (‘Grad’) throughout the dataset, providing no variability.
Received_At_2	Duplicate of Date_of_Contact and may cause confusion.

Renamed Columns for Clarity and Professionalism

Old Column Name	New Column Name	Reason
Name	Outreach Status	Indicates the status of each outreach (e.g., Sent for Evaluation).
Outcome_1	Outreach Outcomes	Represents the result of each outreach e.g., Missing 0 Transcript (Non-Study Group).
Comments	Transcript Evaluation	Describes the evaluation result (e.g., Evaluation Completed).
Caller_Name	Counselor	Refers to the individuals who reached out to the students.

Final Cleaning Summary

Before Cleaning	After Cleaning
40 Columns	27 Columns
5000 Rows	5000 Rows

Link to Cleaned Dataset
https://docs.google.com/spreadsheets/d/1NxD3QrU0mSoHvNqSsVU_TCBv4uOwTm57/edit?usp=drivesdk&ouid=105515635796612148924&rtpof=true&sd=true

Dashboard documentation

Team Name: SLU 0501 DVT Team 22

Team Members:
Abu Yussif (abuyussif125@gmail.com)
Adeniyi Aishat (aishatadeniyi418@gmail.com)
Ayesha Ashraf (ayeshaashraf015@gmail.com)
Mpfariseni Ephraim (ephraimmpfariseni@gmail.com)
Madhubala Khatri (madhubalakhatri53@gmail.com)

Week: Third(3)

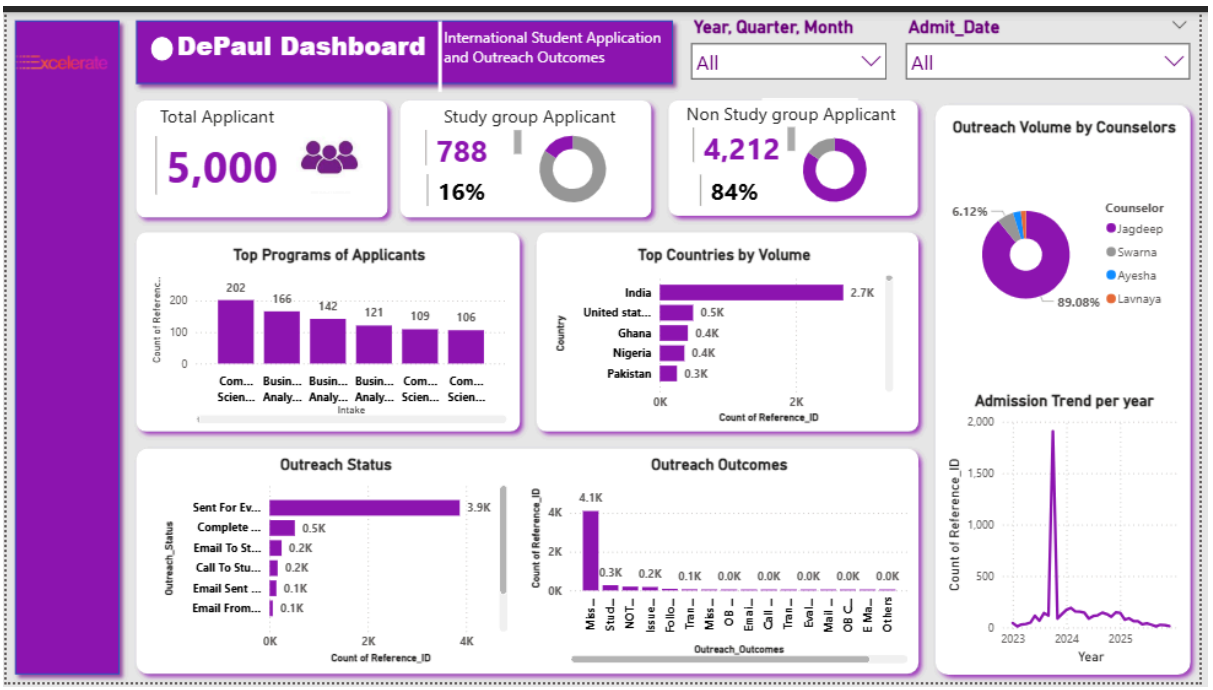
Title:
DePaul Dashboard | International Student Application and Outreach Outcomes

Dataset Link:
https://docs.google.com/spreadsheets/d/1NxD3QrU0mSoHvNqSsVU_TCBv4uOwTm57/edit?usp=drivesdk&ouid=105515635796612148924&rtpof=true&sd=true

Introduction
This report highlights the key details and structure of the DePaul International Student Application and Outreach Outcomes Dashboard.
It covers components such as charts, filters, colors, and data features.
The dashboard provides insights into international student application trends and outreach outcomes for DePaul University.
It enables stakeholders to monitor application progress, counselor performance, and engagement effectiveness.

Dashboard Purpose
The purpose of this dashboard is to visualize international student application data and outreach outcomes for DePaul University.
It helps stakeholders analyze applicant trends, counselor performance, and outreach effectiveness to support data-driven decision-making.

Dashboard



Data Source and Description

Data Attribute	Details
Data Source	Excel File
Dataset Name	Copy_of_Depaul_Data.xlsx
Number of Records	7,543
Number of Columns	80

Data Preparation

1.Removed duplicates and deleted unnecessary columns.
2.Renamed columns for clarity and consistency.
3.Ensured proper formatting of date and categorical data types.
4.Final dataset contains 27 columns and 5,000 rows.

Dashboard Objectives

1.To identify admission performance and trends over the years.
2.To assess the effectiveness of outreach efforts.
3.To monitor outreach status and counselor engagement.
4.To identify trends by country, outreach status, and counselor.

Dashboard Sections

Section	Description
KPIs	Displays 5,000 applicants - 788 (16%) from study groups and 4,212 (84%) from

	non-study groups.
Top Program of Applicants	Shows the top programs applicants applied for, e.g., Computer Science MS 2024 (September 24) emerge.
Outreach Volumes by Counselors	Displays counselor performance and outreach activities/outcomes per counselor.
Application Trends	Visualizes applications over time by country or program
Outreach Outcomes	Highlights key results on outreach such as “Evaluation Completed,” “Missing transcript etc.

Key Metrics and Calculations

Formula / Definition	**Purpose**
COUNT(Reference_ID)	Measures total submitted applications
Calculate(Application_Source = 'Study Group’) Calculate(Application_Source = 'Non Study Group’)	Tracks total applications from study group and non study group sources.
(Applications from Study Group / Total Applications) * 100	Measures the conversion rate from study group sources.
Based on pie chart percentage of outreach activity	Identifies the most active counselor.

Visualizations Used

Visualization Type	**Purpose**
Bar Chart	Compares top countries and outreach outcomes by Reference_ID.
Column Chart	Shows top programs and outreach status effectiveness
Line Chart	Tracks monthly or yearly application trends.
Pie Chart	Displays percentage distribution of application sources and counselor effectiveness.
Table	Provides drill-down analysis by year,

	quarter, month, and day of admission.
KPI Cards	Displays total applications, sources, and conversion rates

Filters
1.Applied filters include:
2.Program (Intake)
3.Country
4.Outreach Status
These filters were necessary due to the numerous categories present in the dataset.

Colour used
The chosen colors were applied to make the dashboard visually appealing, interactive, and easy to interpret. They also ensure consistency and maintain a cohesive visual design throughout the dashboard.

Challenges
1.Intake programs included dates, making it difficult to compare identical programs with different dates.
2.Outreach status contained various counselor messages, leading to incoherent categorizations.
3.The college column included degree information at the front, complicating degree-type analysis.
4.Outreach outcomes had inconsistent terms, making interpretation difficult.

Recommendations
1.Add a column for Admission Status.
2.Create a fixed template for outreach reporting to ensure uniformity.

Conclusion
The dashboard provides a comprehensive overview of DePaul's international outreach and application efforts.
It enables faster evaluation, balanced workload management, and improved communication tracking across the admissions and outreach teams.

Appendix
Link to Dashboard:
https://drive.google.com/file/d/1L7wpas5LSvxPf8Zmlt-Syp7y-4fje29_/view?usp=drivesdk
Link to Demo video of Dashboard:
https://drive.google.com/file/d/1k_tCe6l6Sl65PWDR6DrNJGx_ZBUvs03-/view?usp=drivesdk

Week 3 Insight Report

Team Name: SLU 0501 DVT Team 22

Team Members:
Abu Yussif (abuyussif125@gmail.com)
Adeniyi Aishat (aishatadeniyi418@gmail.com)
Ayesha Ashraf (ayeshaashraf015@gmail.com)
Mpfariseni Ephraim (ephraimmpfariseni@gmail.com)
Madhubala Khatri (madhubalakhatri53@gmail.com)

1.Purpose of the Insight Report

The purpose of this report is to interpret insights derived from the interactive dashboard and translate visual patterns into meaningful, actionable findings. The dashboard focuses on international student applications, outreach performance, admission trends, and counselor effectiveness to support data-driven decision making.

2.Key Insights from the Dashboard

Insight 1: The majority of Applicants Come from Non-Study Group Channels.

Observation

Out of 5,000 total applicants:
84% (4,212) are Non-Study Group applicants
16% (788) are Study Group applicants.

Interpretation

Non-Study Group channels are currently the primary driver of applicant volume. While Study Groups contribute fewer applicants, they may still play a role in targeted or high-quality leads.

Why This Matters

This indicates that outreach strategies outside Study Groups are more scalable and impactful in terms of volume. However, performance quality (conversion, admissions) should be evaluated before deprioritising Study Groups.

Actionable Recommendation

- Maintain Non-Study Group outreach as the main acquisition channel
- Conduct deeper conversion analysis to evaluate Study Group effectiveness beyond volume.

Insight 2: Applications Are Highly Concentrated in Specific Programs

Observation

The “Top Programs of Applicants” chart shows that a small number of programs (e.g. Business Analytics, Computer Science-related programs) account for the highest applicant volumes.

Interpretation
Applicant interest is not evenly distributed across programs. Certain programs consistently attract significantly more interest.

Why This Matters
Understanding program demand helps with:

- Resource planning
- Marketing prioritisation
- Intake capacity planning

Actionable Recommendation
- Focus outreach and marketing spend on high-demand programs
- Review low-demand programs for repositioning or targeted messaging

Insight 3: India Dominates Applicant Country Volume
Observation
India accounts for the largest share of applicants (~2.7K), followed by:
-United States
-Ghana
-Nigeria
-Pakistan

Interpretation
Applicant volume is heavily skewed toward a small number of countries, particularly India.

Why This Matters
Geographic concentration presents both an opportunity and a risk:

- Opportunity: Strong market presence and brand awarenes
- Risk: Over-dependence on a single region

Actionable Recommendation

- Continue strong engagement in India
- Diversify outreach efforts in emerging markets such as Africa and Southeast Asia

Insight 4: Outreach Activity Is High, but Completion Rates Are Uneven

Observation:
The Outreach Status chart indicates a high volume of “Sent for Evaluation” statuses but significantly fewer completed or follow-up activities.

Interpretation:

Many applications enter the evaluation stage but do not progress efficiently through subsequent phases.

Why This Matters:
This suggests possible bottlenecks in:
-Follow-up communication
-Document verification
-Applicant guidance
-Actionable Recommendations:
-Strengthen follow-up workflows after evaluation.
-Introduce automated reminders or checklist systems to assist applicants.

Insight 5: Outreach Outcomes Are Skewed Toward Initial Contact

Observation:
The Outreach Outcomes chart shows a large number of “Missing” or “Not Responded” cases and very few finalised outcomes.

Interpretation:
Initial contact often fails to convert into meaningful engagement or resolution.

Why This Matters:
Low outcome completion indicates inefficiencies in outreach strategy or message relevance.

Actionable Recommendations:
-Refine outreach content and scheduling.
-Implement structured follow-up cadences (2–3 touchpoints per lead).
-Track conversion metrics beyond outreach volume.

Insight 6: Counselor Performance Is Unevenly Distributed

Observation:
The Outreach Volume by Counselors chart shows one counselor handling a disproportionate share of the outreach workload.

Interpretation:
Outreach responsibilities and performance are not evenly distributed among counselors.

Why This Matters:
This may cause:
-Counselor fatigue or burnout
-Uneven applicant experience
-Inefficient resource use

Actionable Recommendations:
-Rebalance outreach assignments among counselors.
-Document and share best practices from top performers.

Insight 7: Admission Trends Show a Sharp Spike in 2024

Observation:
The Admission Trend by Year chart reveals a significant spike in 2024, followed by a stabilisation period.

Interpretation:
This may be linked to enhanced outreach activities, policy adjustments, or increased market interest during that cycle.

Why This Matters:
Understanding the drivers behind this surge can help replicate the success in future recruitment years.

Actionable Recommendations:
-Investigate 2024-specific outreach or admission strategies.
-Apply lessons learned to optimise future intakes.

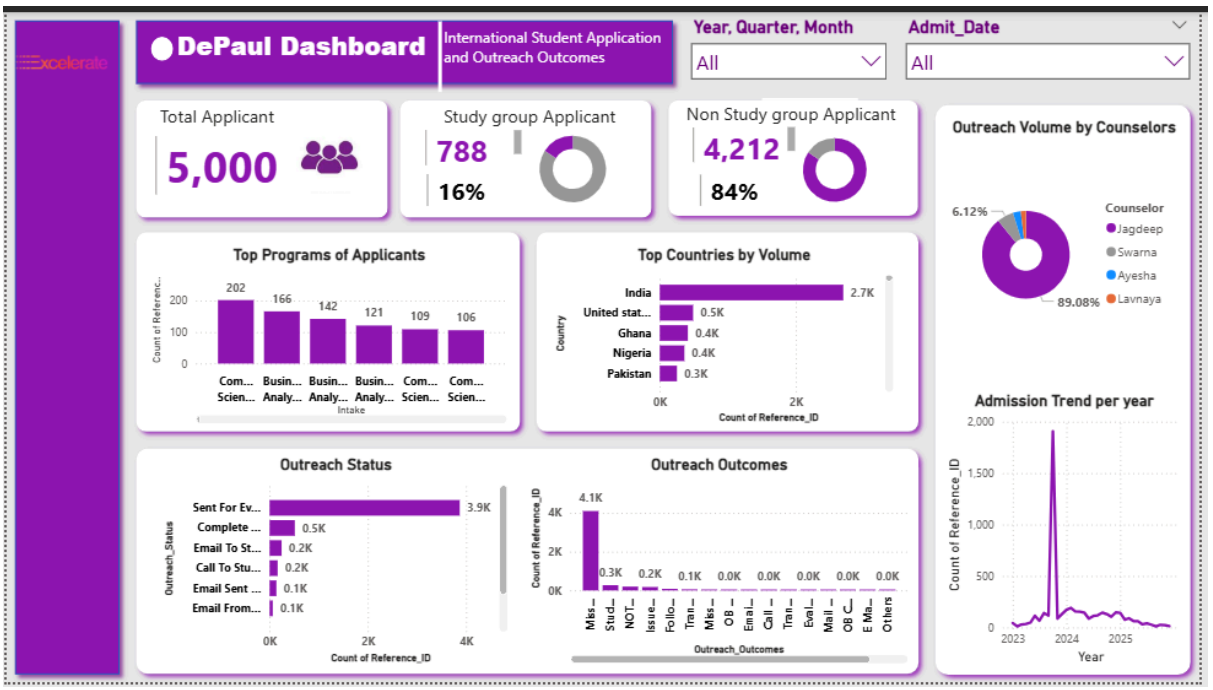
3. Overall Strategic Recommendations

Based on all insights:
1.Prioritise high-performing programs and regions while expanding into new markets.
2.Shift focus from outreach volume to outreach effectiveness.
3.Strengthen follow-up and mid-funnel engagement processes.
4.Balance counselor workload to ensure consistent applicant experience.
5.Leverage historical trend peaks to plan future outreach timing.

4. Conclusion

The DePaul International Student Application and Outreach Dashboard provides a clear, data-driven view of application trends, outreach performance, and counselor efficiency. While outreach activity is robust, opportunities exist to improve engagement quality, follow-up consistency, and conversion effectiveness. Implementing the recommended strategies will enhance overall recruitment efficiency and lead to more sustainable, impactful outreach outcomes.

5. Dashboard


DePaul Dashboard
International Student Application and Outreach Outcomes
Year, Quarter, Month
All
Admit_Date
All
Total Applicant
5,000
Study group Applicant
788
16%
Non Study group Applicant
4,212
84%
Outreach Volume by Counselors
6.12%
89.08%
Counselor
Jagdeep
Swarna
Ayesha
Lavnaya
Top Programs of Applicants
Count of Referenc...
202
166
142
121
109
106
Com... Scien...
Busin... Analy...
Busin... Analy...
Busin... Analy...
Com... Scien...
Com... Scien...
Intake
Top Countries by Volume
Country
India
2.7K
United stat...
0.5K
Ghana
0.4K
Nigeria
0.4K
Pakistan
0.3K
Count of Reference_ID
Admission Trend per year
Count of Reference_ID
2,000
1,500
1,000
500
0
2023
2024
2025
Year
Outreach Status
Outreach_Status
Sent For Ev...
3.9K
Complete ...
0.5K
Email To St...
0.2K
Call To Stu...
0.2K
Email Sent ...
0.1K
Email From...
0.1K
Count of Reference_ID
Outreach Outcomes
Count of Reference_ID
4.1K
0.3K
0.2K
0.1K
0.0K
Miss
Stud
NOT
Issue
Follo
Trai
Miss
OB
Emai
Call
Trai
Eval
Mail
OB C
E Ma
Others
Outreach_Outcomes